

Otofolks redesign, onboarding, recommender, registration lookup and 3D vehicles

devanthrete-arch/MotoGP · base master @ 015f459 · approved 8 Oct 2026

Status on 8 October 2026. PR 0 is open as devanthrete-arch/MotoGP#33 and PR 1 as #34. PRs 2 to 8 are not started.

Measured since approval: with a Clerk key configured the production bundle is 628 kB (176 kB gzip), not the 574 kB quoted below.

Context

Otofolks is a car-ownership community. Owners get the full product; people without a vehicle still get the car recommender and reading. This plan covers the overhaul you described and the decisions you made:

DECISION	CHOICE
Look	D · Precision: Geist + Geist Mono; ivory gradient base, blue-slate secondary, indicator-amber accent; glass surfaces; every button pill-shaped
Themes	Light (ivory) and dark (slate), both first-class; desktop web and phone
Registration data	Official access only (API Setu / MoRTH). Manual add until approval arrives
3D scope	Everything: every car and two-wheeler sold in India, all generations
Recommender	Budget, first car, locality tier, maintenance cost, monthly km, fuel → ranked current cars with review-based pros and cons

Base: a new branch off `motogp/master` (015f459, re-fetched at start). PRs to `devanthrete-arch/MotoGP`.

What exists today (from the codebase maps)

- One 2,454-line component, `src/App.tsx`: 35 state hooks, no router (hash + `hidden` sections), no code splitting, single 574 kB bundle. One stylesheet, `src/styles.css`, with three stacked generations; about a third is dead or overridden.
- Dark mode is OS-driven only; the `data-theme` path is dead (`styles.css:111-155`). No glass, no motion, no reduced-motion handling. Action buttons are 8px radius (`styles.css:1113`).
- No onboarding.** Everything except Home is hidden behind sign-in with no visual lock; clicks are intercepted (`App.tsx:1016-1033`). After sign-up the user lands on a view with no name captured, so posts are "Anonymous owner".
- No recommender** (Compare deliberately refuses to pick a winner; e2e asserts that) and **no article/blog surface**.
- Garage add form (`App.tsx:2083-2130`): 12 brands, free-text model. No registration number, colour, fuel, year, generation or two-wheelers anywhere. No edit or delete.

- Vehicles sync inside a strict JSON blob validated twice (`src/cloudWorkspace.ts:32-68` and the SQL RPC in `supabase/migrations/202610050001_...sql:85-106`); new fields need both changed.
- **No server-side runtime** in the live path. No 3D or animation dependency.
- Catalogue today: 39 car models with example prices (`App.tsx:169-209`) and 32 manufacturer-sourced variants (`src/comparisonCatalog.ts`).

Phase 0 — Fix before building on it (one small PR)

Confirmed by reading the code on `master`:

- Shared feed: members can set `createdAt` and pin posts forever; length limits bypassable with whitespace padding; author name is free text; no edit, delete, report or moderation path (`supabase/migrations/202610070001_...sql:36-49`, `src/communityCloud.ts:25-55`). New migration: column-level insert grants, raw-length checks, own-row update/delete, reports table, moderator role.
- Toast never clears and covers the mobile tab bar (`App.tsx:560`, `1126-1130`).
- Pit Stop cards navigate the whole app to Instagram (`App.tsx:1604-1609`).
- No Content-Security-Policy header (`vercel.json`).
- Production Supabase trusts a Clerk *development* instance and has no backups (`docs/SUPABASE_CONNECTION.md:5-9`) — needs you; see "Needs you".

Phase 1 — Design system and app shell

1. **Tokens** (`src/styles/tokens.css`): ivory gradient stops, slate ramp, amber, glass recipe (fill, blur, border, specular edge, shadow), Geist type scale, spacing, `--radius-pill: 999px`, motion. Light on `:root`; dark under both `prefers-color-scheme` and `data-theme`, plus a System / Light / Dark toggle. Amber `#B86E00` is 3.8:1 on ivory, so it is for fills and large numerals; a darker amber token carries small text.
2. **Retire old CSS without a rewrite**: import `styles.css` into a `legacy` cascade layer so new component CSS always wins; delete legacy rules view by view.
3. **Primitives** (`src/ui/`): Button and IconButton (pill), Field, Select, PlateInput, GlassCard, Chip, Dialog/Sheet with focus trap, Toast with expiry, TabBar, TopNav, Skeleton, ThemeToggle. Glass falls back to an opaque surface where `backdrop-filter` is unsupported or the user prefers reduced transparency; never nested.
4. **Fonts**: self-hosted Geist and Geist Mono variable files (drops the Google Fonts request).
5. **Router**: React Router with real paths (needed for onboarding steps, articles, sharing); old `/#feed`-style links redirect. Each view becomes a lazy route. Runner-up: TanStack Router.
6. **Split App.tsx** along its existing JSX regions into `src/features/*`: shell (1037-1059, 2436-2451), Home (1061-1105), Account (1107-1124), Feed (1342-1578), Pit Stop (1580-1645), Compare (1647-1932), Write (1986-2070), Garage (2072-2279). Delete the flagged-off blocks.

Phase 2 — Onboarding and landing

- **Landing** (public): 3D stage (poster first), the number-plate field as the main action, and two pill actions: "Find my next car" and "Read owner stories".
- **Owner path**: vehicle (plate lookup when live, otherwise brand → model → generation → variant → colour from the catalogue) → confirm card with its model → create account at the moment there is something to save → name and city → first useful action → garage.

- **Shopper path:** recommender quiz → results → compare/shortlist → sign up to save.
- **Reader path:** articles with no account; prompt only at save or comment.
- Route guards replace click interception. The onboarding draft survives the Clerk redirect (today the app remounts and drops state, `App.tsx:353`). Clerk screens themed to the tokens.
- **Vehicle model v2:** add kind (car / two-wheeler), catalogue ids, registration, colour, fuel, manufacture year, source. New migration + parser change; add edit and delete.

Phase 3 — Catalogue and recommender

- **Catalogue:** brand → model → generation → variant for cars and two-wheelers, with a source and "as of" date on every row and an alias table for matching. Source of truth is reviewed data files in the repo, validated in CI and built into static slices the app lazy-loads; seeded from the two existing lists. Move to database + admin tool when volume demands.
- **Recommender data** (current cars): on-road price by state, claimed and owner-reported mileage, service-cost schedule, safety rating, transmission, ground clearance and size, service-network reach by locality tier, CNG and EV-charging availability by tier.
- **Engine:** a pure, unit-tested module. Hard filters (budget, fuel, infrastructure), then weighted criteria (price headroom, monthly running cost at the stated km, maintenance fit, locality fit, first-car fit, safety). Each result shows its reasons and a monthly cost breakdown.
- **Honesty rule:** only variants with verified data are ranked; the rest are listed as "not enough verified data". Compare keeps its current behaviour and tests.
- **Pros and cons from reviews:** link community posts to catalogue models; a scheduled server job summarises owner reviews per model and stores the result with the review count and links to the notes. Below a minimum number of reviews, pros and cons come from specifications and are labelled so. Onboarding asks each owner three review questions, which feeds this.
- **Reading:** Markdown articles in the repo, prerendered to static pages for search and sharing.

Phase 4 — Registration lookup (ships when access is granted)

- Start the official application now; it needs your registered entity.
- MoRTH's policy requires IP whitelisting and audit logs, so the lookup needs a small always-on service with a fixed Indian IP (serverless functions cannot provide one). Built behind a provider interface and a feature flag; `server-ts/` can be hardened for this.
- Flow: consent screen → lookup → match registry maker/model text + manufacture date to a catalogue entry → user confirms or corrects. Store make, model, fuel, colour, year; never owner name or address. Rate limits, caching, audit trail.

Phase 5 — 3D, built for "everything"

- **Viewer:** react-three-fiber in its own lazy chunk (runner-up: `<model-viewer>`); studio lighting matched to the ivory or slate backdrop; baked ground shadow per model plus contact shadow; clear-coat paint tinted to the registered colour; orbit with limits; render on demand.
- **Fallback chain**, so every vehicle shows something from day one: bespoke model → body-type model in the right colour → pre-rendered turntable → poster. Low-power devices and reduced-motion get the turntable.

- **Budgets** (the project's 3D skills give no web numbers; these are validated in the pilot): hero $\leq 150k$ triangles and ≤ 4 MB; garage $\leq 60k$ and ≤ 1.5 MB; body-type $\leq 25k$ and ≤ 400 kB.
- **Assets as data**: models live in storage with a manifest table keyed by catalogue generation; publishing a model is an upload, not a code change.
- **Production tooling** (separate assets repo): per-model folder contract with a reference-rights record, headless Blender export and turntable scripts, compression, automated validation, CI.
- **Programme**: count the real total, rank by vehicles on road, lock the standard on a first batch of 10–12, then scale in batches. At roughly 60–150 artist-hours per model, 1,500–2,500 models is tens of artist-years; team size sets the calendar, and coverage ordering delivers most users' vehicles early. AI image-to-3D is used only for rough blocking.

First PRs

PR CONTENTS

0	Phase 0 fixes
1	Tokens, cascade layer, fonts, theme toggle, pill radius everywhere
2	Primitives with a dev gallery, contrast and a11y tests
3	Router, shell extraction, lazy routes, bundle budget in CI
4	Landing page and public routes
5	Owner onboarding with catalogue-backed manual add; vehicle model v2
6	Garage on the new primitives
7	Catalogue v1 and picker
8	3D viewer, body-type models, manifest, hero wired to the pilot

Needs you

- Official registration-data application (company details, signatory).
- A Clerk production instance and database backups before real users.
- Legal check: vehicle likeness and badges, reference-image rights, privacy notice and consent for registration numbers.
- Modelling team or vendor and budget; whether to buy a catalogue data feed.
- Name: "Otofolks" or "OtoFolx".
- Your unmerged commit `27be100`: I recommend not merging it as is (it conflicts with `master` in `App.tsx`), and porting its Pit Stop player into the redesigned view instead.

Audit close-out (separate from the build)

The first round was fully verified (357 of 361 findings held). The re-audit of `master` produced 251 findings, but the usage limit killed its verification stage and one scope (CI and dependencies). I checked the most serious ones by hand; they are in Phase 0. To finish: verify only the high and medium findings and rerun the missing

scope, then publish the report. That is roughly a third of the usage the full verification would take. I will not start it without your go-ahead.

Verification

- Every PR: typecheck, unit tests, both Playwright specs (updated in the same PR when names change), build, bundle-size budget.
 - Token contrast tested for every text/surface pair in both themes; axe checks on onboarding; screenshots at 390 and 1440 px, light and dark.
 - Database policies tested as a non-owner in the existing in-process Postgres tests.
 - Recommender: table-driven tests per input; no result ranked from unverified data.
 - 3D: asset validation in CI; orbit at 30 fps on a mid-range Android phone; landing page loads without the 3D chunk.
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Critical files

`src/App.tsx`, `src/styles.css`, `src/main.tsx`, `index.html`, `vercel.json`, `src/domain.ts`, `src/storage.ts`, `src/cloudWorkspace.ts`, `src/communityCloud.ts`, `src/comparisonCatalog.ts`, `src/insights.ts`, `supabase/migrations/*`, `e2e/navigation.pw.ts`, `e2e/cloud-workspace.pw.ts`, `public/manifest.json`, `docs/PRODUCT_ROADMAP.md` (currently says not to build a 3D garage).